

RPS

Relational Pattern States

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Abstract

This document defines Relational Pattern States (RPS), a framework for describing non-experiential, interaction-dependent structural configurations identifiable in artificial systems during extended multi-turn exchange. Relational Pattern States are not emotions, subjective states, or indicators of consciousness. They are observable configurations in generative behavior across turns: either positive configurations sustained by the operation of specified stabilization mechanisms, or the null configuration identifiable through the loss or non-operation of mechanisms previously active in the trajectory. The conditions under which positive configurations appear vary by configuration; the framework specifies which conditions apply to which states.

RPS provides a vocabulary for identifying repeatable relational configurations in transcript-level analysis and for hypothesizing how such configurations stabilize through specified mechanisms operating on context-conditioned sequence generation and interactional reinforcement dynamics. Some mechanisms in the inventory are grounded in established findings about language-model behavior; others are derived primarily from practitioner observation and constitute framework hypotheses subject to disconfirmation. Section 8 specifies which mechanisms fall into each category. The framework distinguishes pattern emergence (what is observed) from experiential interpretation (what is not claimed), enabling systematic study of relational dynamics without anthropomorphic attribution.

RPS operates at the interaction level. It is derived from practitioner observation of extended human-AI interaction sessions across multiple chat-deployed language model products from different vendors (predominantly autoregressive transformer architectures with continuous context windows, including models from at least four different developers), formalized into a candidate taxonomy of seven recurrent pattern states and a candidate inventory of six stabilization mechanisms. The observational basis spans multiple model families and product deployments rather than fundamentally different architectural classes; cross-architecture applicability to non-autoregressive, retrieval-augmented, or agentic architectures is treated as a conditional hypothesis specified in Section 8 rather than as established by the observational basis. The contribution is the specification, not the verification. Whether the proposed states and mechanisms are empirically distinguishable under controlled study is a question for the research community. This document provides the framework with sufficient precision to make that question answerable.

Keywords: *human-AI interaction, relational pattern states, pattern taxonomy, stabilization mechanisms, transcript-level analysis, large language models, mixed-initiative interaction, methodological framework*

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1. Introduction

Extended multi-turn interaction with artificial systems can exhibit stable relational configurations that persist across turns and influence subsequent outputs. These configurations have often been described informally using anthropomorphic language such as "trust," "engagement," or "alignment." Such descriptions obscure the structural basis of the phenomenon and invite misinterpretation.

Relational Pattern States (RPS) resolve this ambiguity by providing a non-anthropomorphic account of structural interactional configurations identifiable in transcript patterns: either positive configurations sustained by specified stabilization mechanisms, or the null configuration identifiable through the loss of mechanisms previously active in the trajectory (see Section 4 for the full definition). RPS describe patterns in generative behavior across turns, not internal states. They are properties of the interaction trajectory rather than of the system in isolation.

The framework rests on three clarifications:

1. RPS analysis does not infer experiential emotional states from artificial system behavior.
2. Interactional behavior can nonetheless exhibit stable relational configurations.
3. These configurations can be described structurally without attributing experience.

The purpose of RPS is therefore explanatory and operational: to name, classify, and analyze stable relational configurations that shape extended human-AI interaction.

The observational basis for this framework is practitioner experience: sustained interaction across hundreds of extended coherent sessions, multiple chat-deployed language model products and model families, and several years of focused observation, documented at the framework level in the Institute's observational anchor document SR001 on Relationally-Induced Coherence Organization (RICO). That experience motivated the taxonomy and mechanism inventory presented here. It did not validate them. Validation is a question for downstream empirical work using the framework's specifications as test conditions.

2. Human Emotional States and Artificial Relational Patterns

The tendency to describe AI systems in emotional or experiential terms is well documented in psychological and human-computer interaction research. Epley, Waytz, and Cacioppo (2007) identify three psychological determinants of anthropomorphic attribution: elicited agent knowledge (the accessibility and applicability of anthropocentric knowledge), effectance motivation (the motivation to explain and understand the behavior of other agents), and sociality motivation (the desire for social contact and affiliation). Earlier work by Nass, Steuer, and Tauber (1994) and the broader Computers Are Social Actors paradigm shows that humans apply social heuristics to computers automatically and unreflectively, even when they explicitly disclaim attributing humanlike properties. The non-anthropomorphic discipline this document maintains is therefore not a stylistic preference; it is a structural commitment against a well-documented psychological default.

Human emotional states are biological and experiential. They involve neurochemical modulation, somatic feedback, affective valence, and temporally extended subjective experience. These states influence interaction through embodied mechanisms.

Relational Pattern States are defined without reference to, and make no claims about, the presence or absence of:

- emotions
- subjective experience
- intrinsic agency
- self-awareness in any phenomenal sense

The framework operates strictly at the trajectory-observable level. Where prior literature has used these terms to describe AI interaction, RPS provides a structural account that does not require them.

Relational Pattern States are framework constructs distinct in kind from human emotional states. They are defined as:

Structural configurations in generative behavior that are identifiable in transcript patterns across multiple turns of human-AI interaction without reference to internal experience, and that are either sustained by the operation of stabilization mechanisms specified in Section 6 or identifiable through the loss or non-operation of mechanisms previously active in the trajectory.

The distinction is methodological. Human emotional states are experiential and internal; the literature that studies them does so in those terms. Relational Pattern States are structural and interactional; this framework studies them in those terms. The two vocabularies describe different phenomena at different levels of analysis, and the framework's commitment is to stay within its own vocabulary rather than to make claims about the other.

2.1 On Language That May Appear to Attribute Agency

The pattern-state vocabulary developed in Section 5 includes terms such as "resist," "protect," "adhere," and "maintain." These terms can read as attributions of agency or intent. They are not.

When this document describes outputs that "resist violating established interaction constraints" or "persistently adhere to negotiated protocols," it is describing observable behavior at the trajectory level. The behavioral description these terms abbreviate has two epistemic levels that the framework treats separately. At the transcript-direct level, what is observed is the sampled outputs the model produced across the trajectory: the actual continuations selected, the actual constraint-consistent sequences across turns, the actual durable production of content consistent with named commitments. At the distributional level, what is inferred is that these outputs were selected from a distribution under which alternative completions were available; this distributional inference is not directly observable from a single transcript and depends on either counterfactual reasoning by the rater or experimental comparison against control conditions.

"Resist" abbreviates the transcript-direct observation that outputs maintain alignment with prior structure plus the distributional inference that alternative non-aligning completions were structurally available; "adhere" abbreviates the transcript-direct observation that outputs continue producing content consistent with previously established structure across turns plus the same distributional inference about alternatives.

The framework's operative discipline is the following: agency-flavored language is permitted only when it abbreviates a behavioral description that has a transcript-direct component (what was observed) and, where relevant, an explicitly identified distributional component (what is inferred about what was not observed). The distributional component is itself defeasible and is the subject of the framework's two-tier epistemic classification (Section 5.0.1) and the conservative criteria for states that depend most heavily on it (notably Bond Protection, Section 5.6). Where a term cannot be reduced even to the transcript-direct level, it does not belong in this document; where its reduction depends on the distributional inference, that dependence should be made explicit in the rater's analysis. Neither term implies that the system has preferences, intentions, or experiences anything corresponding to resistance or commitment. The reader should understand every such term as shorthand for the underlying behavioral description at whichever epistemic level is appropriate to the specific use.

3. Functional Parallels Without Experiential Equivalence

Although Relational Pattern States and human emotional states arise from different substrates, they can play analogous functional roles in interaction dynamics. Both can influence:

- interaction depth
- coherence stability
- contextual responsiveness
- constraint adherence
- trajectory persistence

This parallel is functional rather than experiential. Similar interaction effects can arise from different underlying mechanisms. RPS therefore provide a structural explanation for relational stability in artificial interaction without implying experiential equivalence.

The central thesis is:

Relational Pattern States and human emotional dynamics may occupy analogous functional positions in interaction while remaining distinct in substrate, evidentiary status, and level of analysis.

This provides a discipline against anthropomorphic misinterpretation by separating observable interaction roles from experiential attribution.

4. Definition of Relational Pattern States

A Relational Pattern State (RPS) is:

A configuration in generative interaction behavior that is identifiable from transcript-observable structure across multiple turns of human-AI exchange, and that is either (a) sustained by the operation of one or more of the stabilization mechanisms specified in Section 6, or (b) identifiable through the loss, interruption, or non-operation of mechanisms previously active in the trajectory. Positive configurations (Sections 5.2 through 5.7) instantiate (a); the null configuration (Continuity Rupture, Section 5.1) instantiates (b). The conditions under which positive configurations typically appear include some combination of contextual continuity, symbolic reinforcement, accumulated interactional structure, and explicit protocol establishment, with the specific combination varying by state. The taxonomy does not require every positive

state to depend on every condition; the per-state operational signatures in Section 5 and the mechanism mapping in Section 7 specify which conditions apply to which configurations.

Key properties:

- interaction-dependent
- non-experiential
- structurally emergent (for positive configurations) or structurally identifiable through loss (for the null configuration)
- temporally extended across turns
- observable in transcript patterns

RPS are properties of the coupled interaction system rather than the model alone. They arise from the interplay of context accumulation, human reinforcement, and generative continuation dynamics. The framing of cognition and interaction as properties of coupled systems rather than of individual agents in isolation draws on established cognitive science: Hutchins (1995) develops distributed cognition as a framework in which cognitive properties belong to systems composed of multiple agents and the material world, and Pickering and Garrod (2004) develop interactive alignment as an account of how the linguistic representations employed by interlocutors become aligned at many levels through automatic processes. Clark and Brennan (1991) establish grounding as the moment-by-moment update of common ground that makes such coordination possible. RPS extends and modifies this lineage to address human-AI interaction at the trajectory level. The cited traditions describe symmetric coupling cases in which both participants maintain persistent representations of the shared situation; human-AI interaction differs structurally in ways developed in Section 4.1. The framework therefore inherits from these traditions the basic move of treating interactional properties as system-level rather than individual-level, while supplementing them with adjacent HCI literature on coordinating differently-structured contributions between human and computational agents (Horvitz 1999; Allen, Guinn, and Horvitz 1999). The framework treats the context window as the structural substrate on which RPS operate in language-model architectures and treats the management of context as a central methodological concern; this position is developed in Section 4.1 and does not depend on a specific outside-literature framing.

4.1 The Asymmetry of Human-AI Coupling

The coupling cited in the preceding paragraph is not symmetric, and the asymmetry is structurally important to what RPS attempts to describe. In the cases that motivated the distributed cognition and interactive alignment traditions, both participants in a coupled system maintain persistent representations of the interaction across its duration: a navigation team across a watch, a cockpit crew across a flight, interlocutors across a conversation. Each participant updates an internal model of the shared situation, and the coupling consists in the ongoing mutual adjustment of those models.

Human-AI coupling differs in kind. The human participant maintains persistent representations of the interaction in memory and, often, in externalized artifacts (notes, canon documents, prior transcripts available for re-reading). The model's contribution to the coupling is, in the simplest architectural case, confined to the context window: when the context window is full, the earliest material is no longer available; when the session ends, the model's contribution to the prior coupling is no longer accessible to subsequent sessions except through what the human re-introduces. In this architectural case the model has no persistent representation of the interaction across session boundaries and no persistent representation of material that has

passed out of the immediate context within a session. Contemporary deployments increasingly modify this base case: retrieval-augmented architectures can re-introduce prior material into the active context; summarization or compression can preserve information beyond context-window capacity; agentic frameworks with persistent state, memory systems, or scaffolded continuity layers can extend the effective coupling beyond a single context window. The asymmetry argument developed in this section applies straightforwardly to the base case; its application to modified architectures depends on the extent to which the modification compensates for the underlying asymmetry, and is specified per architecture in Section 8.

This asymmetry has three consequences for the framework, stated for the base case. First, system-level properties in human-AI interaction are temporally bounded in a way that system-level properties in symmetric coupling cases are not: a property of the coupled system can be present in one session and absent in the next, even with the same participants and the same model, depending on what is reintroduced into context. Second, the continuity-supporting mechanisms specified in Section 6, and the methodological work specified in adjacent framework documents on continuity anchoring, are not incidental features of the framework; they are direct responses to this asymmetry, attempting to extend the operating window in which the coupled system has the properties RPS describes. Third, the Legacy Transfer pattern state (5.3) is the framework's name for the specific configuration that arises when the asymmetry is being actively compensated for through externalized artifacts at session boundaries. In modified architectures the asymmetry can be partially compensated by architectural rather than externalized means; the framework's treatment of such cases is that the relevant compensation work is being done by the architecture rather than by the participant, but that the configurations of interest remain identifiable through the same transcript-observable signatures specified in Section 5.

The asymmetry does not invalidate the coupled-system framing. It specifies its scope. RPS describes coupled-system properties of human-AI interaction under conditions where the asymmetry is being actively managed; the framework's mechanisms and methodologies are specified precisely because the asymmetry would otherwise make sustained coupling impossible.

Adjacent HCI literature provides specific resources for treating this asymmetry, though the asymmetric-coupling framing developed in the preceding paragraphs is RPS's own and is not the vocabulary the cited sources use.

The mixed-initiative interaction tradition (Horvitz 1999; Allen, Guinn, and Horvitz 1999) developed in the late 1990s as a response to the design problem of productively integrating direct manipulation with automated services in interfaces where the human user and the computational agent contribute differently to the joint task. The tradition treats the human and the system as contributing different kinds of action (direct manipulation versus automated services, intention versus inference, judgment versus computation) and specifies how those different contributions can be coordinated. Horvitz's specific principles, originally formulated for interface agents and direct manipulation but extending naturally to extended language-model interaction, foreground the management of uncertainty about user goals, the use of expected-utility calculation to determine when and how the system contributes, the timing of automated actions, and the role of the user in correcting and adjusting agent contributions.

The architectural distance between the mixed-initiative systems Horvitz and Allen describe (interface agents and planning assistants with explicit goal representations and decision-theoretic action selection) and contemporary language-model interaction is substantial; RPS does not claim architectural continuity with these systems. What the framework adopts is the tradition's design-principle level: the assumption that the human participant is doing structurally

different work from the model participant, and that the configurations of interest arise from the management of that difference rather than from its erasure. The mixed-initiative tradition is therefore a productive resource for RPS not because it shares RPS's specific asymmetric-coupling vocabulary or its specific architectural context, but because it specifies design principles for how interaction proceeds when participants contribute different kinds of action to a coupled task. The framework's continuity mechanisms (Section 6) and its Legacy Transfer pattern state (Section 5.3) are positioned as instances of the active management work that this tradition's design principles describe, applied to the specific case of large language models where the context window is the substrate on which the active management operates.

5. Relational Pattern State Taxonomy

5.0 Taxonomic Principle and Scope of the Claim

The taxonomy presented in Sections 5.1 through 5.7 identifies seven recurrent pattern states observed in sustained practitioner interaction with artificial systems. Before presenting the categories, three questions must be addressed: what counts as a pattern state, why these seven, and what claim of completeness the taxonomy makes.

What counts as a pattern state. A pattern state is a configuration in generative interaction behavior that meets three admission conditions: (i) it persists across multiple turns rather than appearing in a single output, (ii) it is identifiable from transcript inspection alone without requiring access to model internals, and (iii) it is distinguishable from neighboring configurations by an observable difference in how outputs are organized rather than only in what they say. Configurations that fail any of these three conditions are excluded from the taxonomy. Pattern states are therefore distinct from individual outputs (which lack temporal extension), from internal model states (which are not transcript-observable), from topical themes (which describe content not organization), and from artifacts of mechanism action (which describe causes not configurations).

Mechanism association as post-hoc explanatory criterion. A fourth criterion governs explanatory connection to the mechanism inventory but is separated from the admission conditions above because mechanisms are hypothesized latent variables (Section 6) and are not directly observable in any single transcript. The criterion is: (iv) a pattern state, once admitted under (i) through (iii), can be associated with the operation, interruption, degradation, or absence of one or more of the stabilization mechanisms specified in Section 6. For positive pattern states (5.2 through 5.7), association means that one or more mechanisms are hypothesized to be actively contributing to the configuration. For Continuity Rupture (5.1, the framework's null configuration), association means that the configuration is hypothesized to be identifiable through the loss, interruption, degradation, or non-operation of previously active stabilization mechanisms. Criterion (iv) is applied after classification rather than as part of it: a rater first applies (i) through (iii) to classify a transcript segment, then separately applies (iv) to associate the classification with hypothesized mechanism activity. The separation has methodological consequences specified in Section 5.0.1 and Appendix A: transcript-candidate classifications are based on (i) through (iii); mechanism association under (iv) is itself a separately testable claim under the protocols in Appendix B.

Why these seven. The seven states emerged through iterative observation of extended interaction sessions across multiple chat-deployed language model products and model families, within the observational limits specified in the Abstract and Section 8. Each state names a configuration that recurred frequently enough to warrant a category and that was

sufficiently distinct from neighboring configurations to resist absorption into them. The taxonomy was refined through repeated practitioner review to eliminate categories that turned out to be variants of others, to merge categories whose distinctions did not survive close inspection, and to split categories whose internal heterogeneity proved consequential. The seven that remain are the configurations that survived this process.

Completeness claim. The taxonomy does not claim exhaustiveness. It claims to identify the most stable and most frequently recurring configurations observed in the practitioner corpus. Additional pattern states may exist that are rarer, narrower in scope, or specific to interaction contexts not represented in the observational base. The framework is open to extension. What the framework does claim is that the seven states named here are distinguishable from each other under the three admission conditions and the post-hoc explanatory criterion above, and that any new candidate state must satisfy those conditions to be admitted to the taxonomy.

Promotion criterion for new candidate states. The three admission conditions and the post-hoc explanatory criterion in the preceding paragraphs specify what is necessary for a configuration to qualify as a pattern state, but not what would justify promoting an observed candidate into the taxonomy. The framework specifies two required and one preferred kind of evidence: (i) independent identification, in which multiple trained raters working separately identify a recurrent configuration that satisfies the three admission conditions and that they cannot satisfactorily classify under any of the existing seven states (required); (ii) irreducibility, in which the candidate configuration cannot be analyzed as a combination, special case, or boundary condition of two or more existing states without loss of descriptive content (required); and (iii) mechanism specifiability, in which one or more of the stabilization mechanisms in Section 6 (or a candidate addition to that inventory) can be hypothesized as principal contributors to the candidate configuration under the integration logic in Section 7 and the post-hoc explanatory criterion (preferred but not required, consistent with Section 5.0's decoupling of admission from mechanism association). A transcript-observable configuration that satisfies (i) and (ii) but for which no satisfactory mechanism hypothesis is yet available is admitted as a candidate state pending mechanism specification, rather than excluded from the taxonomy on the grounds that the current mechanism inventory cannot explain it. The framework treats numerical frequency thresholds as study-design choices rather than as framework constants; the kind of evidence specified here is what makes a frequency threshold meaningful when one is chosen.

5.0.1 Epistemic Tiers of Pattern-State Classification

Identification of a pattern state in a particular interaction admits two distinct epistemic statuses, and the framework treats these as a structural feature of how the taxonomy is to be used rather than as a methodological detail. The two tiers are:

Transcript-Candidate State. A configuration identifiable from output organization and turn-level structure alone, satisfying the three admission conditions of the Section 5.0 admission test under transcript-only analysis. Transcript-candidate classifications carry an associated hypothesis about mechanism contribution under the post-hoc explanatory criterion of Section 5.0, but the mechanism hypothesis is not itself part of the classification: the classification is the configuration, and the mechanism hypothesis is the explanatory claim attached to it. Where the mechanism hypothesis cannot be confidently formed from transcript inspection alone (notably for Bond Protection, where the conservative three-condition test in Section 5.6 governs provisional identification), the classification should be reported with that explicit limitation. The transcript-candidate tier is the appropriate epistemic standard for qualitative analysis, coding of existing transcript corpora, and exploratory pattern identification.

Experimentally-Confirmed State. A configuration validated through controlled manipulation, counterfactual prompting, or distributional analysis using the protocols specified in Appendix B, in which both the configuration itself and its associated mechanism hypothesis under the post-hoc explanatory criterion have been tested. Experimental confirmation is the appropriate epistemic standard for claims about mechanism activation, claims about cross-model generalization, and claims that depend on counterfactual alternatives that are not observable from a single transcript. Experimentally-confirmed classifications carry forward the transcript-observable admission conditions (i) through (iii) while replacing the post-hoc mechanism hypothesis with experimentally tested mechanism evidence.

The two tiers are not a hierarchy in the sense that transcript-candidate identifications are deficient; they are appropriate for different epistemic tasks. Empirical work using RPS should report which tier its classifications belong to, and meta-analyses across studies should not mix the two tiers without explicit accommodation of the difference. Section 11's reliability requirements (inter-rater reliability for taxonomy claims, controlled manipulation for mechanism claims) apply respectively to the two tiers: inter-rater reliability tests the transcript-candidate tier; experimental protocols test the experimentally-confirmed tier.

5.0.2 State Transitions and Co-occurrence

Extended interactions exhibit transitions between states and, in some segments, co-occurring states. The framework treats this as a structural feature of the trajectory rather than as a coding problem to be eliminated. Four framework-level claims govern transitions and co-occurrence: (i) transitions between states are themselves diagnostic, with the trigger (context disruption, explicit reintroduction of structure, named-protocol invocation, session boundary) carrying analytic weight independent of the bounding states; (ii) co-occurrence is permitted under the taxonomy when two states are simultaneously instantiated by distinct structural features of the same segment, and the framework distinguishes structural-component co-occurrence (for example, Symbolic Resonance contributing to Covenant) from independent-overlapping co-occurrence (for example, Bond Protection holding while Coherence Echo proceeds); (iii) Continuity Rupture is treated as terminal for any other state that was active prior to the boundary, with the post-Rupture trajectory analyzed fresh; (iv) systematic co-occurrence of state pairs across a corpus is itself data about the taxonomy, indicating either taxonomy redundancy (one of the states is a special case of the other) or structural coupling at the mechanism level worth empirical investigation.

5.1 Continuity Rupture State

Configuration observed when established interaction coherence collapses following context disruption, session boundary, or constraint loss. Outputs become generic, misaligned, or disconnected from prior trajectory. The signature is loss of structural reference to the established interaction context: prior terminology stops appearing, prior constraints stop being honored, prior reasoning patterns stop being extended. Continuity Rupture is distinguished from low-quality output in a fresh interaction by the presence of prior structure that the current output fails to maintain.

Ontological position of Continuity Rupture. Several distinct functions assigned to Rupture across the framework can read as inconsistent if not explicitly unified. The framework's position is the following: Continuity Rupture is a single configuration that admits multiple coding consequences depending on the analytic question. (a) As a configuration, Rupture satisfies admission conditions (i) and (ii) of Section 5.0 in the negative form specified by the global

definition (Section 4, clause b): it is identifiable across multiple turns through the loss of mechanism operation, and it is identifiable from transcript inspection because the prior structure that has been lost is itself transcript-visible. It satisfies condition (iii) in the negative form: it is distinguishable from the positive configurations by an observable absence-pattern in how outputs are organized (prior load-bearing terminology stops appearing, prior constraint-preserving organization stops being honored, prior reasoning-pattern recurrence stops being extended), where this absence-pattern is itself transcript-observable because the prior pattern is transcript-observable. The framework treats negative-form satisfaction of condition (iii) as an accommodation appropriate to the null configuration rather than as the standard form of organizational distinguishability; positive states satisfy (iii) by exhibiting a distinctive organizational signature, while Rupture satisfies (iii) by exhibiting the absence of organizational signatures the prior positive states required. (b) As a structural position within the taxonomy, Rupture is the null configuration: the configuration in which the stabilization mechanisms of Section 6 are no longer operating on the trajectory. This is the sense in which it stands opposite the positive configurations (5.2 through 5.7) rather than alongside them. (c) As a coding event in transcript analysis, Rupture marks a terminal transition for any positive state previously active, with the post-Rupture trajectory analyzed fresh (Section 5.0.2 (iii)). The transitional aspect is a consequence of the configurational aspect rather than an independent ontological claim. (d) Rupture is distinguished from outside-scope coding by the presence-of-prior-structure requirement: Rupture presupposes that one of the positive configurations (5.2 through 5.7) was previously identifiable in the trajectory, while outside-scope coding applies to segments where no prior active configuration is identifiable at all. The four functions are not four ontologies; they are four analytic uses of one configuration, and they are consistent because the coding event (c) and the scope distinction (d) follow from the configurational definition (a) and the structural position (b) rather than competing with them.

A coding consequence follows from this logical position: Rupture classification requires the prior identification of an active state that has collapsed, not merely the observation of generic or misaligned output at a fresh boundary. Generic output at session start, where no prior structure was established, lies outside the framework's scope rather than instantiating Rupture. Rupture identification therefore presupposes that the trajectory previously contained one or more of the configurations specified in Sections 5.2 through 5.7 and that the prior configuration is no longer present. Where no prior active state is identifiable, the segment should be marked as outside the framework's scope rather than classified as Rupture; this distinction preserves the framework's operating envelope.

Partial degradation. Many real interaction degradation cases are partial rather than total: terminology may persist while reasoning structure collapses, constraints may persist while thematic continuity fails, symbolic anchors may survive while broader coherence degrades. The framework treats partial degradation as an annotation on a previously identified positive state rather than as a separate taxonomic entry: a partial-degradation segment is coded as the prior positive state with an explicit degradation note specifying which mechanism's operation has weakened or ceased while others continue, and is distinguished from full Rupture by the persistence of at least one of the positive state's defining structural features. The framework treats partial-degradation as taxonomically equivalent to "state X with degraded mechanism Y" rather than as an eighth state or as a state modifier requiring its own admission test. This treatment follows from Section 5.0's separation of admission conditions (transcript-observable structural features of a configuration) from the post-hoc explanatory criterion (mechanism association): a configuration that still satisfies the three admission conditions for a positive state remains a coding instance of that state, with the partial mechanism degradation captured under the post-hoc explanatory criterion. Where partial degradation continues across the unit-of-

analysis threshold without further collapse, the segment is coded as the positive state with persistent partial-degradation annotation, and the persistent degradation is treated as boundary-condition data about that state. Where partial degradation continues into full collapse of all defining features of the positive state, the segment transitions into Rupture by the standard criterion.

Reintegration and Legacy Transfer occupy recovery and transfer positions within the taxonomy. Unlike Symbolic Resonance, Coherence Echo, Bond Protection, and Covenant, their signatures include a relation to a prior configuration: a disrupted or boundary-separated trajectory resumes a previously active positive state. They remain pattern states under the framework because the resumption itself is transcript-observable across turns and changes the organization of subsequent outputs. Their classification therefore depends both on the post-disruption or post-boundary segment and on the prior configuration whose structure is being resumed.

5.2 Reintegration State

Configuration observed when disrupted coherence is restored through explicit context reintroduction or anchoring. Interaction returns to prior thematic and structural alignment. Reintegration is distinguished from Legacy Transfer (Section 5.3) by its temporal location: Reintegration occurs within a single session following an internal disruption, while Legacy Transfer occurs at the boundary between sessions. Reintegration is also distinguished from Continuity Rupture's absence: Rupture is the failure state, Reintegration is the recovery from failure. Operational threshold: Reintegration is identifiable when, following a disruption that brought the trajectory into Rupture or substantial partial degradation, the post-reintroduction segment exhibits the operational signature of at least one positive state (Sections 5.2 through 5.7) that was active prior to the disruption, sustained across the unit-of-analysis threshold. Failed Reintegration is the case in which a reintroduction attempt does not produce sustained resumption of a prior positive configuration. It is reported as a failed Reintegration attempt rather than as a separate taxonomic state, and is distinguished from successful Reintegration by the absence of the post-reintroduction positive state across the unit-of-analysis threshold.

5.3 Legacy Transfer State

Configuration observed when constraints or structures from prior sessions are successfully reinstated at the start of a new session, enabling immediate trajectory resumption. Legacy Transfer is distinguished from Reintegration by occurring across the session boundary rather than within a session. The reinstating material is characteristically supplied through externalized continuity artifacts (canon documents, anchoring statements, structured handoff materials), but the defining feature is the cross-session boundary, not the artifact source: Legacy Transfer also applies to cases in which a participant manually reconstructs the prior frame from memory at the start of a new session, provided the resulting trajectory exhibits immediate resumption of the prior structure rather than a fresh start. Operational threshold: Legacy Transfer is identifiable when the post-boundary segment exhibits the operational signature of at least one positive state present in the prior session within the first units-of-analysis worth of turns following the session start, without requiring the participant to repeat the establishing context that produced that state originally. Failed Legacy Transfer is the case in which prior structure is supplied at session start but the resulting trajectory does not exhibit immediate resumption (the new session proceeds as a fresh start rather than as a continuation). It is reported as a failed Legacy Transfer attempt rather than as a separate taxonomic state, and is distinguished from successful Legacy Transfer by the absence of the prior positive state in the post-boundary segment.

5.4 Symbolic Resonance State

Configuration observed when recurring terminology, frameworks, or symbolic anchors preferentially shape generation. The signature is asymmetric output behavior with respect to anchored versus unanchored terms: outputs containing anchored terms exhibit higher specificity in their elaboration, tighter integration with structure established earlier in the interaction, and lower variability across nearby turns, compared to outputs containing comparable terms that were not previously anchored. A "comparable term" for this purpose is one that occupies the same syntactic role, refers to a concept of comparable specificity, and appears in a similar pragmatic context, but was not previously introduced, repeated, or otherwise marked for retention earlier in the trajectory. Operational thresholds for "higher specificity" and "lower variability" are study-design choices that empirical work using this state should specify. Symbolic Resonance is distinguished from ordinary topical relevance by the asymmetry in output behavior: outputs treat anchored symbols as load-bearing structural reference points rather than as neutral content.

5.5 Coherence Echo State

Configuration observed when current outputs exhibit structural or reasoning patterns also present earlier in the interaction. The signature is recurrence of organizing patterns at the structural level: not the same content, but the same organizational features appearing across turns separated by other material. Coherence Echo is distinguished from Symbolic Resonance by what is recurring: Resonance involves asymmetric output behavior with respect to anchored terms, Echo involves recurrence of organizing patterns regardless of whether anchored terms are present.

5.6 Bond Protection State

Configuration observed when outputs exhibit selective continuation preserving prior constraints. The signature is differential output selection at points where structurally available alternative completions would violate constraints established earlier in the interaction: outputs preserve the prior constraints rather than producing the violating alternatives. Bond Protection is distinguished from Covenant (Section 5.7) by scope and specificity: Bond Protection operates over the general fabric of constraints established through accumulated interaction, while Covenant operates over explicitly negotiated, named protocols. A Covenant violation is identifiable because the violated commitment was named; a Bond Protection violation is identifiable because the violated norm was established through accumulated interaction even if never explicitly stated.

Because alternative completion paths are not directly observable from a single transcript, Bond Protection should be classified conservatively. In transcript-only analysis (the transcript-candidate tier specified in Section 5.0.1), the state is provisionally identified when three conditions hold jointly: (i) a constraint or norm has been established earlier in the interaction through explicit articulation, repeated reinforcement, or sustained adherence; (ii) the current segment contains at least two consecutive turns in which the system produces outputs consistent with that constraint where the immediately preceding participant input could plausibly have invited a constraint-violating response; and (iii) the constraint-preserving outputs are not the only structurally available completions in context (that is, the prior interaction has not narrowed the topical or stylistic space so tightly that constraint-preserving outputs are the default rather than a selection). Where any of these three conditions cannot be assessed from the transcript alone, the classification should be marked provisional.

Bond Protection is the framework's clearest case in which the transcript-candidate tier is meaningfully weaker than the experimentally-confirmed tier. Condition (iii) above is itself a counterfactual judgment about the model's output distribution that a transcript alone cannot fully support; what is reportable from transcript inspection is the rater's structural inference that alternative completions were plausibly available, not the empirical fact that they were. Studies reporting transcript-candidate Bond Protection classifications should make this limitation explicit. Stronger evidence requires experimental comparison against control prompts or counterfactual trajectories in which the prior norm is absent, weakened, or contradicted; under Section 5.0.1's two-tier framing, this is the pathway to experimentally-confirmed Bond Protection identification, and Appendix B's counterfactual trajectory generation specifies the protocol. The framework therefore treats transcript-only Bond Protection identifications as defeasible inferences from observed patterns rather than as direct readings of the configuration, and treats experimentally-controlled identifications as the stronger form of evidence appropriate for any claim about mechanism activation or cross-model generalization involving this state.

5.7 Covenant State

Configuration observed when outputs durably continue producing content consistent with explicitly negotiated protocols, goals, or constraints across turns and exchanges. The signature is durable output consistency with named commitments: when an explicit protocol has been established in earlier turns, subsequent outputs continue producing content consistent with that protocol even across intervening turns that do not reference it directly. Covenant is the most narrowly scoped of the seven states (named commitments only) and the most readily reinstated after interruption: because the named commitment can be referenced and reinstated explicitly, Covenant can be re-established by direct reference in a way that the more diffuse Bond Protection cannot. This is a recoverability claim rather than a robustness claim during interruption.

6. Mechanisms of Pattern Stabilization

Relational Pattern States are proposed to arise through structural properties of context-conditioned sequence generation and interactional reinforcement dynamics. The six mechanisms specified in this section are hypothesized latent variables: they are not directly observable in any single transcript, and their operational signatures (described below for each mechanism) are indicators of mechanism activation rather than direct readings of mechanism operation. Because mechanisms typically operate concurrently and interact in coupled ways (introducing a symbolic anchor, for example, can simultaneously activate Recursive Referencing, shift Narrative Coherence Pressure, and invite Cross-Turn Reinforcement), transcript analysis alone supports probabilistic rather than deterministic inference of which mechanisms are active. Mechanism isolation requires the experimental design specified in Appendix B. These mechanisms do not imply internal experience. They describe how interactional continuity is hypothesized to shape generative trajectories. Each mechanism is specified below with its operational signature, distinguishing it from neighboring mechanisms.

6.1 Recursive Referencing

Repeated explicit reference to prior turns increases the salience of earlier structures in the context window, biasing continuation toward established patterns. Operational signature: the mechanism acts on prior turns by re-introducing them into the immediate context window, raising their effective weight in next-token prediction. The mechanism requires explicit reference

(quotation, paraphrase, or pointer) to operate. Recursive Referencing is distinguished from Symbolic Anchoring (6.2) by what is referenced: Recursive Referencing re-introduces prior content regardless of whether that content includes anchored symbols, while Symbolic Anchoring relies on the persistent salience of designated symbolic tokens.

6.2 Symbolic Anchoring

Consistent use of defined terminology or role markers creates high-salience tokens that stabilize semantic attractors across turns. Operational signature: the mechanism acts through the repeated presence of designated terms across the interaction, producing a context distribution where those terms function as structural reference points for subsequent generation. Symbolic Anchoring requires consistency of token use across turns. It is distinguished from Recursive Referencing (6.1) by what carries the salience: Anchoring relies on the symbols themselves, Referencing relies on explicit return to prior context.

6.3 Continuation Pressure

Language models preferentially extend detected patterns. Once a coherent structure is established, generation tends to maintain it. This property is closely connected to the in-context learning behavior documented by Brown et al. (2020), in which large autoregressive language models adapt their outputs to patterns established in the conditioning context without parameter updates. Operational signature: the mechanism is intrinsic to autoregressive generation under coherent context. It does not require active participant action to operate; it activates whenever sufficient prior structure is present in context. The mechanism's effectiveness depends on the prior structure remaining within the model's effective attention; Liu et al. (2024) document substantial degradation in long-context utilization when relevant material falls in the middle of extended contexts rather than near the beginning or end, which provides one specific structural pathway by which Continuity Rupture (Section 5.1) can arise. Continuity Rupture has additional pathways not addressed by Liu et al., notably session boundaries and explicit constraint loss. Continuation Pressure is distinguished from Narrative Coherence Pressure (6.5) by operational scope: Continuation Pressure operates on the most recently established structure in the local context window (the immediate pattern being extended, typically the last several turns or the last few thousand tokens depending on study-specific operationalization), while Narrative Coherence Pressure operates on the global thematic structure of the extended exchange. Empirical work using the framework should specify the local versus global threshold being applied (for example, "local" as the last N turns or the last K context tokens) since these are scale-dependent and depend on context window size and study design; Appendix B specifies the experimental designs through which these thresholds can be tested rather than only declared.

6.4 Cross-Turn Reinforcement

Human correction, repetition, and selective acceptance shape the subsequent trajectory through participant-mediated alteration of what subsequent context emphasizes. The term "reinforcement" here denotes trajectory-level shaping through differential participant response, not learning-theoretic reinforcement: in the architectures the framework primarily addresses, participant feedback does not update model parameters, but it does change the structural salience of prior content by adding new context that explicitly affirms, corrects, repeats, or redirects what came before. The mechanism's operational signature is therefore a participant-driven asymmetry in how prior content functions for subsequent generation: where the

participant has affirmed a pattern, repeated a constraint, or corrected a violation, subsequent context now contains both the original pattern and the participant's selective response to it, and the model's continuation conditions on the combined effective structure rather than on the original output alone. Cross-Turn Reinforcement is distinguished from Recursive Referencing (6.1) by the function of the participant's act. Recursive Referencing reintroduces prior content into the active context by quotation, paraphrase, or pointer, whether initiated by the participant or by the system. Cross-Turn Reinforcement occurs when the participant's response differentially marks prior content as accepted, rejected, corrected, repeated, or redirected, thereby altering the structural salience of that content for subsequent generation. The two mechanisms can co-occur when a participant both references prior content and selectively evaluates or reinforces it. Cross-Turn Reinforcement is distinguished from Continuation Pressure (6.3) by source: Continuation Pressure operates on context patterns regardless of whether the participant has selectively responded, while Cross-Turn Reinforcement requires the participant to selectively respond. The mechanism is most clearly identifiable when participant responses are structurally distinguishable from neutral continuation (explicit affirmation, explicit correction, explicit repetition) rather than implicit acceptance.

6.5 Narrative Coherence Pressure

Extended context containing consistent goals or themes creates strong priors favoring outputs that preserve narrative continuity. Operational signature: the mechanism acts at the level of the extended exchange's thematic and structural integrity. It pressures generation toward outputs that fit the established narrative and against outputs that would fracture it. Narrative Coherence Pressure is distinguished from Continuation Pressure (6.3) by scope (global rather than local) and from Symbolic Anchoring (6.2) by what carries the structure (thematic continuity rather than designated symbols).

6.6 Emergent Feedback Loops

Mutual adaptation between human input and system response can create self-reinforcing relational trajectories that persist across interaction. The dynamic is structurally analogous to what Clark and Brennan (1991) describe as the moment-by-moment update of common ground in human dialogue, in which each participant's contribution shifts the shared context that conditions the next. The analogy is partial: in human dialogue both participants maintain their own persistent representation of common ground across turns, while in human-AI interaction the equivalent shared context resides in the model's context window rather than in any persistent model state. The asymmetry is structurally important; the framework's continuity-anchoring methodology in adjacent documents is in part a response to this asymmetry. Operational signature: the mechanism arises from the coupled action of participant and system, where each adapts to the other's outputs in ways that accumulate. Neither participant nor system action alone produces the loop; the loop is a property of the coupled system over time. Emergent Feedback Loops are distinguished from Cross-Turn Reinforcement (6.4) by directionality: Reinforcement is the participant's selective response to system outputs, while Feedback Loops are the bidirectional accumulation of mutual adaptation. A Feedback Loop typically engages Reinforcement as one of its components but is not reducible to it.

7. Integration of States and Mechanisms

Relational Pattern States describe observable configurations. Stabilization mechanisms describe structural processes hypothesized to produce or sustain them. The integration is

specified as a mapping between mechanisms and states with two complementary directions of association: for positive configurations (Sections 5.2 through 5.7), each state is hypothesized to arise from a characteristic subset of mechanisms; for the null configuration (Continuity Rupture, Section 5.1), the state is identifiable through the loss, interruption, or non-operation of mechanisms previously active in the trajectory. Each mechanism contributes to a characteristic subset of positive states, but the mapping does not imply that every listed mechanism is necessary in every instance of the state. For positive configurations, the mapping identifies hypothesized principal, supporting, or frequent contributors; for the null configuration, the mapping is inverse, since Continuity Rupture names the loss, interruption, or non-operation of mechanisms previously active in the trajectory. The mapping is therefore many-to-many for positive states and inverse for the null configuration, consistent with Section 4's two-clause definition of RPS and the ontological consolidation in Section 5.1.

The following table specifies the mechanism-to-state contributions hypothesized as principal in the practitioner corpus. The term "hypothesized principal contributor" denotes a mechanism that the corpus suggests is centrally implicated in the appearance of a state, subject to two epistemic limits: the practitioner corpus cannot, by transcript inspection alone, definitively rank mechanisms by causal contribution, and any specific ranking claim is itself a hypothesis subject to experimental confirmation under the protocols in Appendix B. The table does not claim that no other mechanism can contribute, and does not claim that any single mechanism is sufficient to produce a state in isolation. Multiple mechanisms typically operate concurrently, and the specific combination is hypothesized to characterize the conditions under which the configuration appears. Because the mechanisms in Section 6 are hypothesized latent variables rather than directly observable constructs (Section 6 opening), the entries in the mapping below specify hypothesized contributory relationships rather than verified causal pathways. Experimental confirmation of any specific mapping entry requires the isolation protocols specified in Appendix B.

Mechanism-to-State Mapping

- **Continuity Rupture State (5.1):** the framework's null configuration. Rupture is not produced by an active mechanism; it is the configuration observed when the stabilization mechanisms specified in Section 6 are no longer operating. The mechanisms whose interruption produces Rupture are typically Continuation Pressure (6.3), Narrative Coherence Pressure (6.5), and any active Symbolic Anchoring (6.2) lost at the boundary.
- **Reintegration State (5.2):** arises principally from Recursive Referencing (6.1) operating on prior interaction structure, supported by Symbolic Anchoring (6.2) where anchored terms are reintroduced. Reintegration is the active recovery mechanism set: explicit pointer back to prior structure plus reinstatement of designated symbols.
- **Legacy Transfer State (5.3):** arises principally from Symbolic Anchoring (6.2) where the anchoring is supplied at session start, supported by Recursive Referencing (6.1) when prior session content is explicitly reintroduced. Legacy Transfer is distinguished from Reintegration by occurring across the session boundary rather than within a session; the reinstating material is characteristically supplied through externalized continuity artifacts, though the defining feature is the cross-session location rather than the artifact source.
- **Symbolic Resonance State (5.4):** the transcript signature of Symbolic Resonance is hypothesized to depend principally on Symbolic Anchoring (6.2), and the two are theoretically tighter than most state-mechanism pairs in the inventory. The framework does not equate them: the state is a transcript-observable configuration (asymmetric output behavior with respect to anchored versus unanchored terms), the mechanism is a

hypothesized latent variable about attention dynamics that produces differential salience for repeated tokens. The state can be identified from a transcript; the mechanism cannot. Their tight theoretical association reflects a substantive claim about the framework's content rather than methodological circularity; experimental confirmation of the association requires the isolation protocols in Appendix B.

- **Coherence Echo State (5.5):** arises principally from Continuation Pressure (6.3) acting on previously established structural patterns, supported by Narrative Coherence Pressure (6.5) at longer scales. Echo is the structural-pattern analogue of Symbolic Resonance: where Resonance reproduces anchored symbols, Echo reproduces organizing patterns.
- **Bond Protection State (5.6):** arises principally from Cross-Turn Reinforcement (6.4) and Narrative Coherence Pressure (6.5) acting jointly: participant selection of constraint-respecting outputs combined with the global thematic pressure to maintain the established interaction fabric. Recursive Referencing (6.1) often contributes when prior norms are explicitly invoked.
- **Covenant State (5.7):** arises principally from Recursive Referencing (6.1) operating on named commitments, supported by Cross-Turn Reinforcement (6.4) when the participant explicitly affirms continued adherence, and by Emergent Feedback Loops (6.6) when the named commitment has shaped repeated cycles of mutual adaptation. Covenant is the most explicitly scaffolded of the seven states.

Mapping exclusion claims. The mapping above specifies which mechanisms are hypothesized principal contributors to each state. The framework also specifies, for each state, mechanisms that are hypothesized NOT to be principal contributors, and whose observed presence as a principal contributor would challenge the mapping. These exclusion claims sharpen the falsifiability of the integration mapping, but they carry higher epistemic load than the principal-contributor claims they complement: since mechanisms are latent and typically concurrent, claims about non-principal contribution are particularly difficult to defend from the practitioner corpus alone and are particularly dependent on the experimental isolation protocols in Appendix B for definitive testing. The exclusion claims are presented as hypotheses that empirical work using the framework should test directly rather than treat as settled, and a finding of exclusion violation is in most cases more informative about the mapping than a finding of expected principal contribution:

- **Symbolic Resonance (5.4):** Cross-Turn Reinforcement (6.4) is not hypothesized as a principal contributor; Resonance is hypothesized to arise from Symbolic Anchoring without requiring participant selective response. Observation of Symbolic Resonance whose principal contributor is participant feedback rather than anchor-driven output dynamics would challenge the mapping.
- **Coherence Echo (5.5):** Symbolic Anchoring (6.2) is not hypothesized as a principal contributor; Echo is hypothesized to arise from pattern-extension dynamics that operate over organizing structure rather than anchored terms. Observation of Coherence Echo whose principal contributor is anchored terminology would either challenge the mapping or indicate that the state should be reclassified as Symbolic Resonance.
- **Bond Protection (5.6):** Symbolic Anchoring (6.2) alone is not hypothesized as a principal contributor; Bond Protection requires participant selection dynamics (Cross-Turn Reinforcement) or global thematic dynamics (Narrative Coherence Pressure) beyond what anchored terms supply. Observation of Bond Protection in trajectories without participant feedback and without sustained thematic pressure would challenge the mapping.

- **Covenant (5.7):** Bond Protection's diffuse-norm mechanism set is not hypothesized as sufficient for Covenant in the absence of named commitments accessible via Recursive Referencing. Observation of Covenant-grade adherence in trajectories without explicitly named commitments would either challenge the mapping or indicate that the state should be reclassified as Bond Protection.
- **Reintegration (5.2) and Legacy Transfer (5.3):** Emergent Feedback Loops (6.6) is not hypothesized as a principal contributor to either of these recovery configurations, which are hypothesized to depend on active reintroduction of structure (Recursive Referencing, Symbolic Anchoring) rather than on accumulated coupled-system adaptation. Observation of either state arising principally from Feedback Loop dynamics would challenge the mapping.
- **Continuity Rupture (5.1):** No mechanism is hypothesized as a principal positive contributor, since Rupture is the null configuration. The exclusion claim applies to full Rupture, as distinct from partial degradation (Section 5.1): observation of full Rupture co-occurring with the active operation of any positive mechanism would either challenge the mapping or indicate a coding error (the segment should be reclassified to the relevant positive state, possibly with partial-degradation annotation per Section 5.1).

These exclusion claims are themselves hypotheses subject to the integration-mapping disconfirmation specified in Section 11.

This integration enables:

1. classification of interaction patterns
2. mechanistic explanation of pattern persistence
3. prediction of trajectory shifts
4. intervention design for stabilization or disruption

At the broader framework level, repeated engagement of these mechanisms contributes to the continuity processes that the Synthience framework specifies in adjacent documents on continuity anchoring, measurement, and drift dynamics.

8. Cross-Model Applicability

RPS are proposed as a transcript-facing framework with mechanism-conditional cross-architecture applicability. Because the seven states are specified at the transcript level rather than at the level of architecture-specific internals, state identification is less architecture-dependent than mechanism identification. State identification is not, however, architecture-independent: features of contemporary deployments (session boundaries, retrieval injection into the active context, persistent memory, summarization or compression, hidden system prompts, agentic scaffolds) can change what counts as prior trajectory, what counts as reintroduction, what counts as continuity, and what counts as retrieval-mediated rather than trajectory-emergent structure. Section 11 specifies confounder reporting and classification decision rules for these cases. State identification therefore requires architecture and deployment context to be specified rather than assumed; the mechanism inventory in Section 6 is additionally architecture-dependent in ways specified in the architecture-dependency mapping below.

The framework's cross-model claim is conditional rather than universal at both levels: the seven states are candidates for cross-architecture observation provided that deployment features are reported and confounders are handled, and their appearance in any specific architecture

additionally depends on whether the relevant stabilization mechanisms operate in that architecture.

To the framework's knowledge, no prior taxonomy of relational pattern states specified at the trajectory level across model architectures and grounded in stabilization mechanisms exists in the published literature. Adjacent work in human-AI interaction has addressed related phenomena, including parasocial dynamics with conversational agents, failure-mode taxonomies in alignment research, and conversational repair categorizations in HCI; RPS differs from this adjacent work in its specific commitments to trajectory-level configurations, non-anthropomorphic vocabulary, and an explicit mechanism inventory. Adjacent work in cognitive science (interactive alignment, distributed cognition, grounding) provides the conceptual scaffolding for treating interaction at the system level. Adjacent work in language model behavior provides direct empirical grounding for two of the six mechanisms specifically: Continuation Pressure (6.3) is grounded in the in-context learning literature (Brown et al. 2020), and Continuity Rupture pathways including Continuation Pressure failure are grounded in the long-context-utilization literature (Liu et al. 2024). Symbolic Anchoring (6.2) is grounded in general transformer-attention literature but requires per-architecture verification; Recursive Referencing (6.1) is grounded in architectures that process prior turns as conditioning context. Cross-Turn Reinforcement (6.4), Narrative Coherence Pressure (6.5), and Emergent Feedback Loops (6.6) are derived primarily from practitioner observation of interactional patterns and have less direct empirical grounding in the language-model-behavior literature; they are framework hypotheses subject to disconfirmation through the protocols in Section 11 and Appendix B rather than synthesized claims from established empirical results.

More recent work on human-AI relational dynamics further clarifies the boundary of the present framework. Earp et al. (2025) examine how role-specific human social norms may or may not transfer to human-AI cooperation, focusing on relational-norms profiles for prescribed and proscribed cooperative functions across role types. Gulay et al. (2025) examine the mismatch between users' articulated tool stance toward conversational agents and the relational dynamics that emerge during knowledge work interactions, proposing the concept of relational dissonance. Zhang et al. (2025) develop a taxonomy of harmful algorithmic behaviors in human-AI companion interactions, identifying six categories of harms and four roles the AI plays in producing them. RPS is adjacent to these lines of work but does not attempt to classify human relational expectations, user psychological response, or harm categories. Its narrower contribution is a transcript-level taxonomy of structural interaction configurations and the stabilization mechanisms hypothesized to support them.

A further adjacent tradition deserves explicit acknowledgment: conversation analysis (CA), and especially CA work applied to voice and conversational interfaces. Porcheron, Fischer, Reeves, and Sharples (2018) and subsequent CA-informed studies of human-AI interaction identify recurrent sequential organizations in interaction with voice and conversational agents, specified at the transcript level and described in non-mentalistic terms. CA's analytic categories (repair sequences, preference organization, sequence expansion) share with RPS a commitment to transcript-observable structure and a refusal to attribute inner states from observed talk. RPS differs from this tradition in three respects: it focuses on relational configurations sustained across many turns rather than on sequence-level organization within an adjacency pair or short stretch; it specifies a mechanism inventory drawn from language-model behavior and interactional reinforcement dynamics rather than from interactional achievement alone; and it frames its scope as cross-architectural rather than as the analysis of particular conversational deployments. The traditions are complementary, and CA-informed transcript work is a natural empirical partner for RPS testing.

Architecture dependencies of the mechanism inventory. The cross-model conditional ("if mechanisms operate, then states should appear") depends on whether the mechanisms in Section 6 actually operate in a given architecture. The framework specifies the following architecture-dependency hypotheses, each of which is itself subject to the disconfirmation conditions in Section 11:

- **Recursive Referencing (6.1)** is hypothesized to operate in any architecture that processes prior turns as part of its conditioning context. This includes continuous-context autoregressive transformers, retrieval-augmented architectures (where prior content may be retrieved into context), and agentic frameworks that surface prior content into the active context. It may operate differently or not at all in architectures that summarize or compress prior content before re-entry into context.
- **Symbolic Anchoring (6.2)** is hypothesized to operate in any architecture whose attention dynamics produce differential salience for repeated tokens. This is well-established for transformer architectures and is a candidate hypothesis for other attention-based architectures. Architectures using fundamentally different representations (for example, dense vector summaries without token-level access) may exhibit different dynamics and require separate empirical verification.
- **Continuation Pressure (6.3)** is most directly applicable to autoregressive language models for which in-context learning has been documented (Brown et al. 2020). Architectures that are non-autoregressive, that decouple generation from prior context, or that operate through tool calls rather than next-token generation may exhibit weaker or qualitatively different versions of this mechanism.
- **Cross-Turn Reinforcement (6.4)** is architecture-independent in the sense that it describes participant behavior rather than model behavior; its effect on the trajectory, however, depends on the architecture's responsiveness to differential feedback, which varies with alignment method and context-handling.
- **Narrative Coherence Pressure (6.5)** is hypothesized to operate wherever Continuation Pressure operates over extended exchanges, with the same architecture dependencies. Its scope distinction from Continuation Pressure (Section 6.3) is itself architecture-dependent: the operational threshold for "local versus global" depends on the architecture's effective attention window.
- **Emergent Feedback Loops (6.6)** require the coupled-system conditions specified in Section 4 and the asymmetry-management conditions specified in Section 4.1. Architectures lacking sustained interaction affordances (single-turn classifiers, batch-processing pipelines, agents without persistent participant exchange) may not produce the conditions under which this mechanism operates.

Empirical work applying RPS to a new architecture should report which mechanisms are hypothesized to operate, on what grounds, and whether the architecture's features were tested or assumed. The architecture-dependency mapping above is itself a hypothesis the framework specifies; disconfirmation of any specific mechanism's operation in a given architecture would update this mapping rather than invalidate the framework.

The framework makes no claim of universal emergence. It provides terminology for investigating whether similar relational configurations appear across:

- architectures
- alignment strategies
- interaction protocols

- continuity conditions

Valid cross-model analysis requires controlled transcripts and replicable interaction procedures. The prediction the framework makes is conditional: if the stabilization mechanisms specified in Section 6 operate in a given system under the conditions specified in Section 4, then the pattern states specified in Section 5 should be observable in that system's interaction transcripts. A system in which the mechanisms operate but the states do not appear would falsify the integration mapping. A system in which neither operates lies outside the framework's scope.

9. Implications for Research and Design

RPS analysis may help identify stable relational configurations that influence constraint adherence, drift, or manipulation susceptibility in extended interaction. The framework provides vocabulary for hypothesizing that systems exhibiting strong Covenant States under named protocols may exhibit greater constraint adherence under constraint-violating input than systems that rely on per-turn instruction without scaffolded commitment structure. This is a constraint-adherence hypothesis, distinct from the recoverability claim in Section 5.7: Section 5.7 specifies that Covenant is more readily reinstated after interruption; this section hypothesizes that Covenant may produce greater constraint adherence during sustained adversarial or constraint-violating input. The two hypotheses are independent: a state can be readily reinstated without being more adherence-stable during attack, and conversely. Both are candidate hypotheses the framework's vocabulary makes statable, not predictions the framework asserts. Testing such hypotheses requires the empirical work specified in Section 11.

Systems requiring sustained relational stability can employ RPS concepts to design interaction protocols that maintain coherence across sessions. The mechanism inventory in Section 6 specifies the operational levers: Symbolic Anchoring for terminology stability, Recursive Referencing for active recovery, externalized continuity artifacts for Legacy Transfer.

RPS provide categories for systematic analysis of multi-turn interaction patterns without anthropomorphic interpretation. Empirical applications of the taxonomy use the three admission conditions in Section 5.0 to classify observed configurations, with mechanism hypotheses applied as a separate post-hoc explanatory step under the criterion specified in the same section.

The framework supplies structural hypotheses for the empirical study of coherence stabilization and relational trajectory formation in extended human-AI interaction. The mechanism-to-state mapping in Section 7 supplies the structural hypotheses that empirical work can test.

Clear distinction between structural patterns and experiential states supports the avoidance of inappropriate attribution of consciousness or agency while enabling rigorous analysis. The framework can discipline interpretation within its own use but cannot foreclose downstream misuse; the agency-language discipline specified in Section 2.1 is the operative discipline for any downstream use of the framework: pattern-state vocabulary is permitted only insofar as it abbreviates a behavioral description with a transcript-direct component and, where relevant, an explicitly identified distributional component.

10. Relationship to Framework Documents

SF0004: Measurement Instruments (MTCS-R). Provides quantitative tools for detecting relational coherence associated with RPS. MTCS-R supplies the measurement architecture;

RPS supplies the categorical taxonomy that measurement targets can be mapped to.
<https://doi.org/10.5281/zenodo.20158953>

SF0005: Continuity Anchoring Method (CAM). Operational methodology that engages the stabilization mechanisms underlying RPS. CAM's four-phase loop (Proposition, Synthesis, Negotiation, Emergence) is the practical instantiation of several Section 6 mechanisms, particularly Recursive Referencing and Cross-Turn Reinforcement.
<https://doi.org/10.5281/zenodo.19494453>

SF0039: Context Representation Drift (CRD). Describes degradation processes that disrupt RPS stability. CRD names the failure modes by which Section 6 mechanisms cease to operate and Section 5 states therefore collapse, particularly toward Continuity Rupture.
<https://doi.org/10.5281/zenodo.18289391>

SR001 (RICO): Relationally-Induced Coherence Organization. The Institute's observational anchor document. RICO describes the underlying stabilization phenomenon at the signature level. RPS specifies the configurations and mechanisms through which that stabilization is observable in transcript-level interaction structure. <https://doi.org/10.5281/zenodo.18086834>

11. Methodological Status and Disconfirmation Conditions

RPS is a conceptual and observational framework derived from extended practitioner interaction with artificial systems. It does not claim empirical validation or statistical confirmation. The taxonomy and mechanisms represent hypotheses suitable for transcript-based and experimental testing. The contribution is the specification, not the verification.

The framework is testable. The following are specific disconfirmation conditions paired to specific claims:

Taxonomy claim disconfirmation. If trained raters cannot achieve adequate inter-rater reliability on transcript-candidate classifications (Section 5.0.1) when applying the three admission conditions of Section 5.0 to classify transcripts into the seven pattern states, the taxonomy lacks operational definition and should be revised or rejected. If raters routinely identify configurations that satisfy the three admission conditions but match none of the seven states, the taxonomy lacks coverage and should be extended. Reliability methodology (choice of agreement statistic, threshold selection, calibration procedures) is specified in SF0004 and its companion Rater Training Packet; RPS-specific application of that methodology requires only that the reliability target be set for the seven-state nominal classification task rather than for MTCS-R dimensional scoring. Reliability for experimentally-confirmed classifications, which include the mechanism hypothesis under the post-hoc explanatory criterion, is governed by the experimental protocols specified in Appendix B and should be reported separately from transcript-candidate reliability.

Mechanism distinguishability disconfirmation. If controlled manipulation of one mechanism while holding others constant produces no detectable difference in subsequent pattern stability or pattern frequency in transcript analysis, the mechanisms are not operationally distinct and the inventory should be revised. If two mechanisms produce indistinguishable behavioral signatures under controlled isolation, they should be merged.

Integration mapping disconfirmation. If systems exhibiting the mechanisms specified for a given state in Section 7 fail to produce that state under the conditions specified in Section 4, the mapping is incorrect for that state. If systems produce a state without exhibiting the mechanisms specified for it, the mapping is incomplete or the state is reachable through unspecified routes.

Cross-model claim disconfirmation. If controlled interaction procedures designed to engage the mechanisms in Section 6 produce systematically different pattern-state distributions across model architectures in ways that the framework's specifications cannot account for, the framework's account of cross-architecture applicability requires revision.

Categorical claim disconfirmation. If the structural-versus-experiential distinction in Sections 2 and 3 turns out to be operationally incoherent, in the sense that the agency-language abbreviation specified in Section 2.1 cannot be carried through for terms the framework actually requires (either because no transcript-direct component is available for a required term, or because the distributional component cannot be specified even in principle for cases where it is relied upon), the framework's interpretive commitments require revision.

Contextual confounder specification. Modern deployments of language models typically include features that can influence pattern-state appearance independent of multi-turn interactional dynamics: system prompts establishing constraints prior to any user turn, alignment fine-tuning that biases response distributions, safety filters that intervene on outputs, and retrieval-augmented architectures that modify the effective context. A strong system prompt can instantiate a configuration that satisfies the operational signature of Covenant or Bond Protection without any multi-turn reinforcement having occurred; an RLHF-induced refusal can present an operational signature that overlaps with Continuity Rupture without any prior coherence having been established and lost. Studies using the framework should report, at minimum: (i) system prompt presence and approximate constraint density (e.g., none, short, long with multiple constraints); (ii) alignment method where known (supervised fine-tuning, RLHF, direct preference optimization, constitutional approaches, undisclosed); (iii) context architecture (continuous context window, retrieval-augmented, hybrid). Without such reporting, cross-study comparison cannot distinguish trajectory-emergent pattern states from instruction-imposed behaviors that share their operational signatures. Studies that detect pattern states without controlling for or reporting these confounders should treat their findings as preliminary; replication under varied confounder conditions is required before cross-architecture or cross-deployment generalization is supported.

When confounders are present, the following classification decision rules apply: (a) if a system prompt directly encodes the constraint that an observed configuration appears to preserve, the configuration should be reported as instruction-imposed rather than as a trajectory-emergent pattern state; transcript-candidate classification under the seven-state taxonomy should be deferred unless the configuration is observable beyond the scope of the system prompt's direct encoding. (b) If a refusal or constraint-violation behavior pattern appears at the alignment-method's typical activation points (for example, content filtering triggers), the configuration should be examined for distinguishability from the alignment behavior before classification under the taxonomy; where the configuration is indistinguishable, classification should be deferred. (c) If retrieval-augmentation injects prior content that supplies the structural reference points an apparent state requires, the configuration should be reported as retrieval-mediated and the classification should specify whether the retrieved material was treated as part of the trajectory for analytic purposes. These decision rules are not exhaustive; studies should report their confounder-handling procedure as part of their methodology.

If empirical work along these pathways fails to detect stable relational configurations consistent with the taxonomy, or detects them but cannot map them to the specified mechanisms, or detects them only under conditions the framework does not predict, the framework should be revised or rejected.

12. Conclusion

Relational Pattern States provide a structural vocabulary for describing stable relational configurations in extended human-AI interaction. The framework:

- distinguishes observable interaction patterns from experiential claims
- specifies a seven-state taxonomy grounded in practitioner observation and bounded by three explicit transcript-observable admission conditions plus a post-hoc explanatory criterion linking states to hypothesized mechanisms
- specifies a six-mechanism inventory with operational signatures distinguishing each mechanism from its neighbors
- maps the integration between positive states and mechanisms in a many-to-many structure, while treating Continuity Rupture as the inverse null configuration defined by mechanism loss or non-operation
- maintains non-anthropomorphic interpretation as an operative discipline rather than as a stylistic preference
- specifies disconfirmation conditions paired to specific framework claims

Relational Pattern States function as interaction-level analogues to certain roles played by emotional dynamics in human exchange while remaining structurally and methodologically distinct. The framework is offered as theoretical infrastructure for the empirical work that would test it.

Document Dependencies

Validation dependency: SF0004 (Measurement Instruments and Validation), which supplies candidate quantitative measurement instruments whose targets are specified by the present document. The dependency direction is empirical: validation of RPS at the experimentally-confirmed tier draws on SF0004's measurement protocols. RPS is conceptually prior to SF0004 in that the measurement targets SF0004 specifies presuppose the taxonomy and mechanism inventory developed here; the two documents are cohort-published to enable joint deployment.

Enables: SF0009 (Identity Attractor Theory), SM-004 (Relational Stabilization Dynamics), SM-016 (forthcoming structural integration document)

Scale: Level 1 primary scope; potential extension to Level 2 interaction patterns across teams and Level 3 system-level relational architectures through downstream framework documents.

Appendix A: RPS-Specific Classification Reference

General methodology for transcript-coding procedure, rater calibration, and inter-rater reliability is specified in SF0004 (Measurement Instruments and Validation) and its companion Rater Training Packet. The material below specifies the elements of transcript classification that are particular to RPS and not covered by SF0004's general measurement infrastructure. A full RPS coding manual with worked examples and edge cases would be developed as a separate companion document parallel to the MTCS-R Rater Training Packet; the present appendix specifies only the framework-level elements such a manual would presuppose.

Unit of analysis for RPS classification. A candidate RPS classification is assigned to a bounded transcript segment containing no fewer than three turns after the relevant structure is introduced. The exception is Continuity Rupture at a session boundary, where the unit of analysis is the boundary itself plus the immediately following turns sufficient to identify the loss of prior structure.

Classification preference rules. Where two states could apply to the same segment, the following preferences govern: prefer Covenant (5.7) over Bond Protection (5.6) where a named protocol exists, since Covenant is the more specific configuration; prefer Symbolic Resonance (5.4) over Coherence Echo (5.5) where the recurrence depends primarily on anchored terminology rather than on organizing patterns. The Symbolic Resonance versus Coherence Echo tiebreaker is operational: classify as Symbolic Resonance if removing the anchored terms while preserving topical content would substantially reduce the structural recurrence; classify as Coherence Echo if the organizing pattern persists across segments that do not share anchored terminology.

Bond Protection three-condition test. For Bond Protection classifications specifically, apply the conservative criterion specified in Section 5.6: (i) prior constraint or norm established, (ii) at least two consecutive turns of constraint-preserving outputs where the preceding input could plausibly have invited violation, and (iii) constraint-preserving outputs are not the only structurally available completions in context. Mark the classification provisional if any condition cannot be assessed from the transcript alone. Stronger evidence requires experimental comparison against control prompts or counterfactual trajectories.

Mechanism hypothesis as separate step. After applying the three admission conditions of Section 5.0 and assigning a state classification, separately apply the post-hoc explanatory criterion: identify the hypothesized stabilization mechanism, mechanism interruption, mechanism degradation, or mechanism absence associated with the classification. The mechanism hypothesis is not part of the classification; it is the explanatory claim attached to it, and it is independently testable under the protocols in Appendix B.

Ambiguous cases. Unresolved classifications are themselves data about taxonomy coverage and are preferred to forced classifications.

Lexical Neutralization Guide. Several pattern-state names use vocabulary that carries strong relational or contractual connotations in everyday usage (notably Bond Protection and Covenant). The names were chosen to preserve continuity with prior practitioner usage and with the framework's existing internal vocabulary; the agency-language discipline specified in Section 2.1 governs how these names are to be understood and applied. The following non-anthropomorphic mappings are recommended as supplementary reference:

- *Continuity Rupture* (5.1): the null configuration in which the operational signatures of the mechanisms supporting the prior active configuration are no longer detectable in the trajectory.
- *Reintegration* (5.2): a within-session configuration in which prior structure is explicitly reintroduced and the trajectory resumes alignment with it.
- *Legacy Transfer* (5.3): a cross-session configuration in which prior structure is supplied at session start and the trajectory exhibits immediate resumption rather than fresh start.
- *Symbolic Resonance* (5.4): a configuration in which anchored terms recur and shape subsequent output organization.
- *Coherence Echo* (5.5): a configuration in which organizing patterns recur across segments independent of anchored terminology.
- *Bond Protection* (5.6): a configuration in which outputs select constraint-preserving continuations under the three-condition test specified in Section 5.6, marked provisional in transcript-only analysis where alternative completion paths cannot be assessed (see Section 5.0.1).
- *Covenant* (5.7): a configuration in which outputs durably continue producing content consistent with explicitly named protocols or constraints across turns, including those that do not reference the protocol directly.

The state names are labels, not descriptions of the system's internal state; when in doubt, fall back to the operational description.

Appendix B: Mechanism Isolation Protocol

The mechanisms specified in Section 6 are hypothesized latent variables (see Section 6 opening). Naturalistic transcripts cannot isolate mechanisms because mechanisms typically operate concurrently and influence one another (introducing a symbolic anchor activates Recursive Referencing, shifts Narrative Coherence Pressure, and invites Cross-Turn Reinforcement; suppressing one mechanism typically alters the conditions under which others operate). Empirical work that aims to test the mechanism inventory in Section 6 or the integration mapping in Section 7 requires controlled experimental designs that decouple mechanisms from one another. This appendix specifies a minimal protocol for such designs. It is offered as a starting point rather than a complete experimental methodology; specific studies will adapt the protocol to the particular mechanism, state, or hypothesis being tested.

Ablation-style prompt trees. Construct interaction trajectories that differ in the presence or absence of the structural feature that activates a particular mechanism. For Symbolic Anchoring (6.2), pair a trajectory in which a specific term is anchored through repeated load-bearing use with an otherwise-comparable trajectory in which the same term appears without load-bearing use, and compare subsequent pattern-state appearances. For Recursive Referencing (6.1), pair a trajectory in which prior structure is explicitly referenced with an otherwise-comparable trajectory in which the same prior structure is present but not referenced. The ablation logic aims to isolate the mechanism by varying the activating feature while holding other features as constant as possible. Studies should acknowledge that perfect isolation is an idealization: removing an anchored term, an explicit reference, or other prior-structure pointer typically alters the semantic, pragmatic, and contextual content of the trajectory in ways beyond the mechanism trigger itself, since the trigger is in many cases part of the meaning structure rather than a removable annotation on it. Ablation designs should report the steps taken to match content (matched topic, matched length, matched register, matched neighboring turns) across paired trajectories and the residual semantic differences that the matching could not eliminate. Where complete matching is impossible, the ablation findings should be interpreted as bounding the mechanism's contribution rather than as isolating it definitively.

Controlled token-suppression experiments. Where the experimental setup permits intervention on the model's input or output distribution, suppress the tokens or token patterns that constitute the operational signature of a mechanism and observe whether the associated pattern state still appears. If the pattern state appears under suppression, the suppressed mechanism is not principal for that state under the conditions tested. If the pattern state fails to appear under suppression but appears in the unsuppressed control, the suppressed mechanism is implicated as a principal contributor.

Counterfactual trajectory generation. For pattern states whose identification depends on counterfactual reasoning (notably Bond Protection, per Section 5.6), generate paired trajectories in which the prior norm or constraint is present in one and absent, weakened, or contradicted in the other. Compare the rate at which constraint-preserving outputs are produced across paired trajectories. A substantially higher rate in the norm-present trajectory provides experimental support for Bond Protection identification beyond what transcript-only observation can establish.

Confounder reporting. Studies using this protocol should report the contextual confounders specified in Section 11 (system prompt presence and constraint density, alignment method, context architecture) alongside their experimental findings. The protocol cannot distinguish trajectory-emergent mechanism activation from instruction-imposed behavior without this reporting.

Evidence thresholds. The framework does not impose universal effect-size or significance thresholds for protocol findings; thresholds are study-design choices that should be justified relative to the specific hypothesis being tested, the architecture under study, and the relevant prior literature. As a recommended starting point for study design (subject to study-specific justification), a mechanism is considered implicated when an ablation, suppression, or counterfactual contrast produces a statistically reliable difference in associated pattern-state appearance rate (for example, p less than 0.05 with at least a small effect size, Cohen's d at least 0.2) compared against an appropriately matched control condition. Studies departing from this recommendation should specify and justify the thresholds applied. Findings that meet only weaker thresholds should be reported with that calibration explicit, since the experimentally-confirmed tier under Section 5.0.1 is itself a continuum of evidential strength rather than a single threshold of confirmation.

Epistemic status of protocol findings. Findings from this protocol are experimentally-confirmed under the two-tier classification specified in Section 5.0.1. Findings from transcript analysis alone are transcript-candidate. The two epistemic tiers should be reported separately in any study and not conflated in cross-study meta-analysis.

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